

S. A. I. Shouborno

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Research interests

Interpretable and physics-informed machine learning for sensor data; wearable health sensing, including cuffless blood pressure and sleep-based cognitive screening; industrial cyber-physical systems; language models grounded in sensor data; efficient on-device inference.

Education

- Ph.D., Electrical and Computer Engineering**, University of Massachusetts Amherst Sep 2026 – May 2028 (expected)
Advisor Bashima Islam, continuing the doctoral work begun at WPI.
- Ph.D. studies, Electrical and Computer Engineering**, Worcester Polytechnic Institute Aug 2023 – Aug 2026
GPA 4.00/4.00.
- M.S., Electrical and Computer Engineering**, Worcester Polytechnic Institute Aug 2023 – Dec 2025
Thesis: Interpretable and Efficient AI for Sensor-Rich Mobile and Embedded Systems.
- B.Sc., Electrical and Electronic Engineering**, Bangladesh University of Engineering and Technology Feb 2016 – Feb 2021
Thesis: Audiovisual emotion recognition using a multi-head attention network, advised by Celia Shahnaz. GPA 3.40/4.00.

Experience

- Research Assistant, University of Massachusetts Amherst**, Amherst, MA Sep 2026 – Present
- Cuffless blood pressure estimation from graphene electronic tattoo sensors, with the Kireev Lab.
 - Fault diagnosis for industrial controllers from sensor and program trace data, with Bosch Research.
- AI for Cyber-Physical Systems Intern, Robert Bosch LLC, Bosch Research**, Pittsburgh, PA May 2025 – Aug 2025
- Built a contrastive multimodal model connecting vision, sensor, program trace and text data for language-guided few-shot functional state classification.
- Research and Teaching Assistant, Worcester Polytechnic Institute**, Worcester, MA Aug 2023 – Aug 2026
- BASH Lab research on LLMs for IMU-based activity reasoning and on budgeted early-exit inference.
 - TA for On-Device Deep Learning (CS 525 / DS 595 / ECE 579), Discrete-Time Signal and System Analysis (ECE 2312), Introduction to Digital Circuit Design (ECE 2029), and Sensors, Circuits, and Systems (ECE 2019).
- Machine Learning Engineer, Hishab Technologies**, Dhaka, Bangladesh Feb 2021 – Jul 2023
- Built telephony conversational AI for Bengali speakers, covering speech recognition, text to speech, NER, language models and grapheme-to-phoneme conversion.
 - Deployed services for banking and microfinance clients, including Bank Asia and Hitachi India.
 - Co-inventor on a U.S. patent application for optimizing interactive voice response sessions.
 - Worked with partner Chowa Giken (Japan) on car interior image enhancement.
- Machine Learning Engineer Intern, Socian Ltd.**, Dhaka, Bangladesh Sep 2020 – Dec 2020
- NLP and speech processing for Bengali, including POS tagging, NER, sentiment and topic classification, and ASR.
- CTO, Gyanjam Corporations**, Dhaka, Bangladesh Apr 2018 – Mar 2019
- Built an e-commerce and e-learning site, designed PCBs, and taught an 8-day C programming workshop for BUET students.
- Research Intern, Onnorokom Electronics**, Dhaka, Bangladesh Jul 2016 – Dec 2017
- Designed science kit products for school students and answered their scientific questions by email.

Publications

12. S. A. I. Shouborno, Emily Ruppel, Sudershan Boovaraghavan, Bashima Islam. **Program Execution Traces Without Instrumentation: Debugging Industrial Controllers from Physical Process Sensors**. AAAI Conference on Innovative Applications of Artificial Intelligence (IAAI-27). *Under review*.
11. S. A. I. Shouborno, Mohammad Nur Hossain Khan, Bashima Islam. **JOLT: Jointly Optimized Layerwise Termination for Budgeted Early-Exit Inference**. IEEE International Conference on Tools with Artificial Intelligence (ICTAI). *Accepted, 2026*.

10. **S. A. I. Shouborno**, Bashima Islam. **Cross-Site Generalization of Interpretable Sleep Features for Predicting Future Cognitive Impairment**. *Computing in Cardiology (CinC)*, 2026.
9. Tyler Bliss, Mahit Verma, Aila Iyer-Singh, Subrata Biswas, **Sheikh Asif Imran**, Bashima Islam. **Concord: An Agreement-Aware Multi-Adjudication Pipeline for LLM Evaluation**. *Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM)*, pp. 502–510, 2026.
8. **S. A. I. Shouborno**, Mohammad Nur Hossain Khan, Subrata Biswas, Bashima Islam. **LLaSA: A Sensor-Aware LLM for Natural Language Reasoning of Human Activity from IMU Data**. *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion)*, pp. 893–899, 2025.
7. T. I. Mahmud, **S. A. Imran**, C. Shahnaz. **Res-SE-ConvNet: A Deep Neural Network for Hypoxemia Severity Prediction for Hospital In-Patients Using Photoplethysmograph Signal**. *IEEE Journal of Translational Engineering in Health and Medicine*, pp. 1–9, 2022.
6. **S. A. Imran**, M. T. Islam, C. Shahnaz, M. T. Islam, O. T. Imam, M. Haque. **COVID-19 mRNA Vaccine Degradation Prediction Using Regularized LSTM Model**. *IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE)*, pp. 328–331, 2020.
5. O. T. Imam, M. Haque, C. Shahnaz, **S. A. Imran**, M. T. Islam, M. T. Islam. **Detection of Tuberculosis from Chest X-Ray Images Based on Modified Inception Deep Neural Network Model**. *IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE)*, pp. 360–363, 2020.
4. A. B. A. Qayyum, K. N. Hassan, A. Anika, M. F. Shadiq, M. M. Rahman, M. T. Islam, **S. A. Imran**, S. Hossain, M. A. Haque. **DOANet: A Deep Dilated Convolutional Neural Network Approach for Search and Rescue with Drone-Embedded Sound Source Localization**. *EURASIP Journal on Audio, Speech, and Music Processing*, pp. 16, 2020.
3. M. T. Islam, **S. A. Imran**, A. Arefeen, M. Hasan, C. Shahnaz. **Source and Camera Independent Ophthalmic Disease Recognition from Fundus Image Using Neural Network**. *IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, pp. 59–63, 2019.
2. A. B. A. A. Qayyum, A. Anika, M. M. M. Miah, M. M. Rahman, K. N. Hasan, M. T. Islam, **S. A. I. Shouborno**, M. F. Shadiq, M. A. Haque. **Direction of Arrival Estimation through Noise Suppression: A Novel Approach Using GSC Beamforming and Room Acoustic Simulation**. *IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, pp. 104–108, 2019.
1. **S. A. I. Shouborno**, T. I. Mahmud, N. Ishraq, R. Ali, T. H. Joy, S. A. Fattah, A. Bhattacharjee, C. Shahnaz. **Complete Automation of an E-commerce System with Internet of Things**. *IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, pp. 81–86, 2019.

Patent

T. Al Muntasir, M. T. Islam, **S. A. Imran**. **System and Method for Optimizing a User Interaction Session within an Interactive Voice Response System**. U.S. Patent Application No. 18/378,118, 2025.

Honors and awards

5th place, IEEE Video and Image Processing Cup at ICIP 2020 (real-time vehicle detection and tracking with fisheye cameras)	2020
4th place, IEEE Signal Processing Cup at ICASSP 2020 (unsupervised anomaly detection for autonomous drones)	2020
2nd place, IEEE Video and Image Processing Cup at ICIP 2019 (activity recognition from body cameras)	2019
National champion and 22nd worldwide, IEEE Xtreme 12.0 programming contest (national champion in 2017 as well)	2018

Service

Chair, IEEE Signal Processing Society student chapter at BUET. Organized the 2019 IEEE SPS Winter School on Multimodal Signal Processing; the chapter grew into the largest in Bangladesh and received the IEEE SPS Student Branch Chapter Growth Reward.	2019 – 2021
IEEE Xtreme ambassador and 2019 country lead for Bangladesh; ran programming workshops.	2017 – 2019

Skills

Machine learning: PyTorch, multimodal and contrastive learning, LLM fine-tuning, early-exit and efficient inference, physics-informed models

Signals: Biomedical signals (PPG, bioimpedance, sleep PSG), IMU, audio and speech, image and video

Systems: SLURM clusters, MLOps, GCP and AWS, Kafka, Jenkins, microcontrollers, FPGA